

Periodic Functions & Modeling

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ 1. I can **graph** basic sine and cosine functions using key features.

RELATED SKILLS

- ☐ I Locate quarter-period key points of a periodic function.
- ☐ D Use tabular patterns to predict graph shape and behavior.
- ☐ A Determine unit-circle coordinates for common angles.

☐ 2. I can **identify** amplitude, midline, period, and phase shift from an equation or graph.

RELATED SKILLS

- ☐ I Determine a sinusoid's midline and vertical shift.
- ☐ I Determine amplitude and vertical reflection.
- ☐ I Determine horizontal shift from a graph or equation.
- ☐ I Determine period from a graph or equation.
- ☐ I Recall and correctly use the terms amplitude, midline, period, phase shift, vertical shift, and asymptote.

☐ 3. I can **graph** transformed sine and cosine functions.

RELATED SKILLS

- ☐ D Distinguish stretches from compressions and horizontal effects from vertical effects.
- ☐ D Locate quarter-period key points of a periodic function.
- ☐ A Interpret transformations applied to function inputs.
- ☐ A Interpret transformations applied to function outputs.

☐ 4. I can **write** sine and cosine equations that represent a periodic graph.

RELATED SKILLS

- ☐ D Connect parameters in a formula to features of its graph.
- ☐ D Determine a sinusoid's midline and vertical shift.
- ☐ D Determine amplitude and vertical reflection.
- ☐ D Determine horizontal shift from a graph or equation.
- ☐ D Determine period from a graph or equation.

☐ **5. I can **construct** a sinusoidal model from contextual information or data.**

RELATED SKILLS

- ☐ I Translate contextual maximum, minimum, cycle, and starting position into model parameters.
 - ☐ A Determine a sinusoid's midline and vertical shift.
 - ☐ A Determine amplitude and vertical reflection.
 - ☐ A Determine period from a graph or equation.
 - ☐ A Restrict a model to meaningful contextual inputs.
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☐ **6. I can **interpret** the parameters and outputs of a sinusoidal model in context.**

RELATED SKILLS

- ☐ D Explain a model parameter in the language of its context.
 - ☐ D Translate contextual maximum, minimum, cycle, and starting position into model parameters.
 - ☐ A Attach and interpret units for rates, inputs, and outputs.
 - ☐ A Identify input and output values in a formula, graph, table, or context.
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☐ **7. I can **graph** tangent, secant, and cosecant functions with their asymptotes.**

RELATED SKILLS

- ☐ I Locate vertical asymptotes of tangent, secant, and cosecant.
 - ☐ D Relate sine/cosecant, cosine/secant, and tangent/cotangent.
 - ☐ A Interpret transformations applied to function inputs.
 - ☐ A Locate quarter-period key points of a periodic function.
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